

Wrangling concepts · reduction & transformation · group_by

Monday, June 1, 2026

 Today: Monday June 1

Before class

- Perusall Assignment: [DC §7.1–7.7](#)

Plan for today

- The wrangling *mindset*: start from the table you want
- The three function families: **reduction**, **transformation**, **data verb**
- First two data verbs: `summarize()` and `group_by()`
- The `na.rm` trap
- The usefulness of `n()` in summaries

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 3](#) · [DC Ch. 7](#) · [dplyr reference](#)

Where we are

For two weeks we've been **making plots**. But that assumed the data was already in the right shape, with one row per glyph and one column per aesthetic.

Tip

Real data almost never arrives glyph-ready. This week is about **getting it there**.

Today we will build the **mental model** that every wrangling function hangs off of.

Reading the pipe

We've learned `|>`. Read it as the word “**then.**”

```
penguins |>           # take penguins, then
  group_by(species) |> #   group it, then
  summarize(n = n())  #   summarize it
```

Note

The *DC* reading writes the pipe as `%>%`. `|>` and `%>%` behave identically, so read them as the same word. We'll use `|>`.

Wrangling mindset

Start from the table you want

DC §7.1 gives the single most useful habit in this whole unit:

Before any coding, sketch the **glyph-ready table** you need, then plan the steps to get there in **plain English**.

For “how does average body mass differ by species?”, the table you *wish* you had is:

species	avg_mass
Adelie	...
Chinstrap	...
Gentoo	...

Here we have one row per species and one column for the number you want. **The case is a species, not a penguin.** That shift in what one row represents is from `summarize()` and `group_by()`.

Plan in English first

DC's smoking example plans the wrangling as four English sentences *before* any code:

1. Make an age-group variable from age.
2. Drop people under 20.

3. Split into groups by age-decade and sex.
4. For each group, count people and count smokers; divide.

Each sentence maps to **one operation**. Today we learn the vocabulary for sentences like #3 (*split into groups*) and #4 (*for each group, count / average*).

A framework for wrangling

Three kinds of building blocks

DC §7.2 boils all of wrangling down to three shapes of data...

- A **data frame**: rows (cases) and columns (variables)
- A **variable**: one column
- A **scalar**: a single value

and three families of functions, sorted by **what goes in and what comes out**:

Family	Input → Output	Examples
Reduction	variable → scalar	<code>mean()</code> , <code>sum()</code> , <code>n()</code> , <code>median()</code> , <code>sd()</code> , <code>n_distinct()</code>
Transformation	variable(s) → variable	<code>x / y</code> , <code>log10()</code> , <code>round()</code> , <code>ifelse()</code> , <code><</code> , <code>==</code> , <code>%in%</code>
Data verb	data frame → data frame	<code>summarize()</code> , <code>group_by()</code> , <code>filter()</code> , <code>mutate()</code> , ...

Reduction functions: many → one

A **reduction** takes a whole column and crushes it down to a single number (DC §7.3).

```
mean(penguins$body_mass_g, na.rm = TRUE) # one number out
```

```
[1] 4201.754
```

We've used many of these before: `mean()`, `sum()`, `median()`, `sd()`, `min()`, `max()`. Here are two new ones:

- `n()` — how many cases (takes **no** arguments)
- `n_distinct()` — how many *different* values

 Note

`n()` doesn't look at any column, it simply counts rows. It only works *inside* a data verb like `summarize()`, not on its own.

Transformation functions: column to column

A **transformation** runs once *per row* and hands back a column the same length as the input (*DC* §7.4).

```
head(penguins$body_mass_g / 1000) # grams to kg, one value per row
```

```
[1] 3.75 3.80 3.25 NA 3.45 3.65
```

Comparisons are transformations too, they return a column of TRUE/FALSE:

```
head(penguins$body_mass_g > 4000)
```

```
[1] FALSE FALSE FALSE NA FALSE FALSE
```

 Tip

Reduction vs. transformation in one question: does it give back *one* number, or *one per row*? `mean()` to one. `round()` to one per row.

Data verbs: data frame to data frame

A **data verb** takes a whole data frame and returns a whole (new) data frame (*DC* §7.5). Here are three rules every dplyr verb obeys:

1. The **first argument is always a data frame**.
2. The other arguments name **columns**, unquoted.
3. The **output is always a new data frame**; the input is never modified.

```
penguins |> summarize(avg = mean(body_mass_g, na.rm = TRUE))  
# penguins itself is unchanged; you get a NEW data frame back
```

Warning

Because verbs never modify their input, `penguins |> summarize(...)` on its own just **prints**. To keep a result, assign it: `out <- penguins |> summarize(...)`.

Each step = one verb + helpers

Each step in data wrangling involves a **data verb** and one or more **reduction or transformation functions**.

The verb is the *action on the table*; the reduction/transformation is the *detail of what to compute*.

```
penguins |>
  summarize(avg_mass = mean(body_mass_g, na.rm = TRUE))
#           data verb      reduction function
```

summarize()

summarize(): collapse a table to a summary

`summarize()` turns many rows into **one row**, using reduction functions.

```
penguins |>
  summarize(
    n      = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE),
    max_mass = max(body_mass_g, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 3
   n avg_mass max_mass
<int>   <dbl>   <int>
1   344   4202.    6300
```

Things to notice:

- Each **named argument** becomes a **column** in the output.
- The output is a **data frame** — one row here, but still a data frame.

The `na.rm` trap

Watch what happens without `na.rm = TRUE`:

```
penguins |>
  summarize(avg_mass = mean(body_mass_g))
```

```
# A tibble: 1 x 1
  avg_mass
  <dbl>
1      NA
```

One missing value anywhere in the column makes the **whole answer** NA. The reduction can't average around a hole it doesn't know how to fill.

Warning

This is the AI failure we dissected on Friday: code that runs, looks fine, and quietly returns NA. The fix is `na.rm = TRUE`, but the *skill* is noticing the NA in the first place.

`group_by()`

`group_by()`: summarize, but per group

`group_by()` doesn't compute anything itself. It **tags** the data frame so the *next* verb runs once per group.

```
penguins |>
  group_by(species) |>
  summarize(
    n      = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE)
  )
```

```
# A tibble: 3 x 3
  species      n avg_mass
  <fct>    <int>   <dbl>
1 Adelie   152    3701.
2 Chinstrap  68    3733.
3 Gentoo  124    5076.
```

Same `summarize()` as before, now **one row per species** instead of one row total.

i Note

`group_by()` should (almost) always be followed by another verb. On its own it just attaches an invisible label, the work happens in the verb after it.

Grouping by more than one variable

List several columns to get one row per **combination**:

```
penguins |>
  group_by(species, sex) |>
  summarize(
    n      = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE)
  )
```

```
# A tibble: 8 x 4
# Groups:   species [3]
  species sex      n avg_mass
  <fct>   <fct> <int>   <dbl>
1 Adelie female   73   3369.
2 Adelie male     73   4043.
3 Adelie <NA>      6   3540
4 Chinstrap female  34   3527.
5 Chinstrap male    34   3939.
6 Gentoo  female   58   4680.
7 Gentoo  male    61   5485.
8 Gentoo  <NA>     5   4588.
```

Two things worth a pause:

- There's an NA row for `sex` indicating some penguins have missing sex. `group_by()` keeps them as their own group. (Dropping them needs `filter()`.)
- You may see a message about the output being “*regrouped*”. This is harmless for us, and we can add `.groups = "drop"` to silence it.

How group_by() changes the case

Wrangling changes **what one row means**.

Table 1: group_by() resets the meaning of a case.

Pipeline	One row =
penguins	one penguin
... > summarize(n = n())	the whole dataset
... > group_by(species) > summarize(...)	one species
... > group_by(species, sex) > summarize(...)	one species × sex combo

Tip

Before you run a `group_by() |> summarize()`, predict **how many rows** the result has. It's the number of distinct group combinations. If your prediction is wrong, you've learned something about your data.

Trust, but verify

Always include an n()

Average mass by the finest grouping:

```
penguins |>
  group_by(species, island, sex) |>
  summarize(
    n      = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE),
    .groups = "drop"
  )
```

```
# A tibble: 13 x 5
  species island sex      n avg_mass
  <fct>   <fct> <fct> <int>   <dbl>
1 Adelie  Biscoe female    22  3369.
2 Adelie  Biscoe male     22  4050
3 Adelie  Dream  female    27  3344.
```


4	Adelie	Dream	male	28	4046.
5	Adelie	Dream	<NA>	1	2975
6	Adelie	Torgersen	female	24	3396.
7	Adelie	Torgersen	male	23	4035.
8	Adelie	Torgersen	<NA>	5	3681.
9	Chinstrap	Dream	female	34	3527.
10	Chinstrap	Dream	male	34	3939.
11	Gentoo	Biscoe	female	58	4680.
12	Gentoo	Biscoe	male	61	5485.
13	Gentoo	Biscoe	<NA>	5	4588.

Some groups have a handful of birds, others have dozens. **A mean from a group of 3 is not the same kind of fact as a mean from a group of 60.**

Good habit

Warning

Whenever you compute a group summary, **add `n = n()`**. A surprising-looking average is very often just a tiny group. Without the count, you can't tell a real effect from noise.

This is also a verification move, exactly like Friday: the number that *looks* like an answer might be resting on almost no data.

Tomorrow

What's next

Today: the framework + `summarize()` + `group_by()`. That's already enough to answer a lot of “how many / what's the average, by group” questions.

Tomorrow (*DC* Ch. 10) we add the four verbs:

- `filter()` — keep some **rows** (e.g. drop the NA sexes)
- `select()` — keep some **columns**
- `mutate()` — add a **new column** (this is where transformations live)
- `arrange()` — **reorder** rows

With the two we covered today, these handle “take a messy CSV and produce a useful summary.”

What we covered today

- **The mindset:** sketch the glyph-ready table, plan in English, one sentence per operation
- **The framework:** reduction (variable→scalar), transformation (variable→variable), data verb (frame→frame)
- **summarize():** many rows to one; named args become columns; output is a data frame
- **na.rm = TRUE:** one NA poisons a whole reduction; spotting it is the skill
- **group_by():** tags the frame so the next verb runs per group; changes what a “case” means
- **n():** include it in every summary so you never trust a mean from three rows

Activity

Activity: summarising the penguins

Roughly 25 minutes. Individually or with a neighbor.

- [Activity Canvas Link](#)

You’ll classify functions into the three families, build up `summarize()` and `group_by()` pipelines on `penguins`, deliberately trip the `na.rm` trap, and practice predicting the **shape** of a grouped result before you run it.

To turn in: save your `.qmd` and rendered HTML.

Wrap-up & looking ahead

Tomorrow (Tuesday): the row and column verbs, `filter`, `select`, `mutate`, `arrange`, the rest of the dplyr toolkit.

Upcoming Deadlines

- **Tue 6/2, before class:** [DC §10.1–10.6](#)
- **Thu 6/4, 11:59 p.m.:** [Project: Team formation](#)
- **Fri 6/5, 11:59 p.m.:** [Homework 3: Data Wrangling & Reshaping](#)

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 3](#) · [DC Ch. 7](#) · [dplyr reference](#)